ANIKET KULKARNI

VLSI Design & Technology | RTL Design | Verification | Physical Design

Pune, Maharashtra | Expected graduation: 2027

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PROFESSIONAL SUMMARY

Final-year VLSI Design & Technology student. I build processors, ALUs, and memory blocks in Verilog, test them with directed simulations, and study the synthesis and layout reports to see what still needs work. My largest project is a five-stage RV32IM + Zicsr microcontroller SoC.

TECHNICAL SKILLS

RTL & verification: Verilog, SystemVerilog, Icarus Verilog, VCS/Verdi, directed testbenches, waveform debugging

Architecture: RISC-V RV32IM + Zicsr, MIPS-style CPUs, pipelining, FIFOs, fixed-point Q7.8, CSR/traps, SPI

Implementation: Yosys, OpenLane/SKY130, OpenROAD/OpenSTA, KLayout, Magic/Netgen; reviewing Design Compiler and IC Compiler II reports

SELECTED VLSI PROJECTS

RV32IM + Zicsr Microcontroller SoC | OpenLane / SKY130 | Oct 2026

- Built a five-stage CPU with forwarding, a 32-cycle multiply/divide unit, machine-mode traps, and UART, GPIO, timer, and SPI peripherals. Added synchronous WFI sleep; directed firmware simulation passed.
- Ran the phase-5 design through OpenLane on a 1 mm² die (21,584 routed cells). Saved reports show zero Magic DRC and Netgen LVS errors; extracted timing and antenna issues still need work.
- Compared phase-4 and phase-5 OpenSTA reports. Estimated typical-corner switching power dropped from 9.65 to 6.93 mW (28.2%) after sleep-control and layout changes.

Five-Stage MIPS-Subset Processor | Verilog RTL | 2026

- Built a 32-bit pipeline with forwarding, load-use stalls, and branch/jump flushing. Directed tests checked six registers and one memory value; generic Yosys synthesis passed.

4 × 4 Fixed-Point Matrix Multiplier | Verilog RTL | 2026

- Built a sign-magnitude Q7.8 datapath with 64 multipliers and 48 adders. The testbench checked all 16 outputs for one matrix pair, and generic Yosys synthesis passed.

16 × 32 Synchronous FIFO | Verilog RTL | 2026

- Wrote parameterized and structural single-clock versions. Eight directed test suites cover data movement, reset, full/empty flags, threshold, and boundary cases.

Other Verilog work | 16-Bit CPU, 8-Bit Datapath, 4-Bit ALU | 2026

- Also built a custom-ISA processor, an externally instructed ALU/register datapath, and a registered eight-operation ALU. Source and directed simulations are available for each.

RELEVANT EXPERIENCE

Digital Electronics & VLSI Intern | Codec Technologies Pvt. Ltd. | Jul–Aug 2026

- Worked through a two-month training program in digital logic, number systems and codes, and VLSI fundamentals.

Technical Assistant Intern | Dynomerk Controls India Pvt. Ltd. | Jan–Feb 2026

- Helped assemble industrial control panels and work with electronic components. Used AutoCAD and automation test tools for Engine and ABS systems.

EDUCATION

Undergraduate Degree in VLSI Design & Technology | APCOER, Pune | 2023–2027

Anantrao Pawar College of Engineering and Research | Savitribai Phule Pune University

WORKSHOPS, CERTIFICATIONS & TRAINING

Semiconductors for AI & AI for Semiconductors | SemiX AI SIG, IIT Bombay | 17 Oct 2026

In a guided workshop, studied a 4 × 4 systolic-array reference design (16 PEs), followed its skewed dataflow and golden-model checks, and reviewed a scripted Synopsys flow from synthesis to routing.

- 4th Annual SemiX Summit, IIT Bombay, 14–17 Oct 2026: semiconductor entrepreneurship, AI compute, and power electronics.
- Physical Design Using Synopsys — VLSIGuru, Sep 2026; VLSI Specialist Skillset Test — Beep, Aug 2026.
- Semiconductor Manufacturing — CeNSE, IISc, Apr 2026; RTL to GDSII — SemiconTantra, Feb 2026.
- Verilog HDL Fundamentals — Udemy, Oct 2025; AI PC Workshop — SemiX, IIT Bombay, Sep 2025.

LEADERSHIP

Technical Symposium Coordinator, SEMICONIX – APCOER: co-organized quizzes and presentations with the student team.